

Wykaż, że $\frac{1+2\cos 88^\circ \cdot \cos 2^\circ}{\cos^2 2^\circ - \cos 88^\circ \cdot \sin 2^\circ} = \frac{1+\operatorname{tg} 2^\circ}{1-\operatorname{tg} 2^\circ}$.

① przekształcamy lewą stronę równania

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$L = \frac{1+2\cos 88^\circ \cdot \cos 2^\circ}{\cos^2 2^\circ - \cos 88^\circ \cdot \sin 2^\circ} = \frac{\overset{a^2+b^2}{1+2\overset{2ab}{\sin 2^\circ \cos 2^\circ}}}{\cos^2 2^\circ - \sin^2 2^\circ} = \frac{\sin^2 2^\circ + 2\sin 2^\circ \cos 2^\circ + \cos^2 2^\circ}{\cos^2 2^\circ - \sin^2 2^\circ} =$$

$\cos 88^\circ = \cos(90^\circ - 2^\circ) = \sin 2^\circ$

$$= \frac{(\sin 2^\circ + \cos 2^\circ)^2}{(\cos 2^\circ - \sin 2^\circ)(\cos 2^\circ + \sin 2^\circ)} = \frac{\sin 2^\circ + \cos 2^\circ}{\cos 2^\circ - \sin 2^\circ} = \frac{\frac{\sin 2^\circ}{\cos 2^\circ} + \frac{\cos 2^\circ}{\cos 2^\circ}}{\frac{\cos 2^\circ}{\cos 2^\circ} - \frac{\sin 2^\circ}{\cos 2^\circ}} = \frac{\operatorname{tg} 2^\circ + 1}{1 - \operatorname{tg} 2^\circ} = P$$

c. k. d.